KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 19 · SCIENCE

Rock Layers and Fossils: Reading Earth's History

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 18):** Name Earth's four spheres. Then give one example of two spheres interacting.

2. What is a fossil? Write what you think it is and one place you might find one.

3. Predict: In a stack of rock layers, which layer do you think is the oldest? Why?

4. Draw a cliff with three rock layers. Draw a fossil in one layer and label which layer you think formed first.

5. What is one question you have about fossils or the Grand Canyon?

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Read the Rock Record

Use evidence from today's lesson. Write in complete sentences.

1. Undisturbed layers, top to bottom: W (sandstone, tracks), X (limestone, corals), Y (shale, leaf prints), Z (sandstone, ripple marks). List them oldest to youngest. What rule did you use?

2. For layers X and Y above, what environment does each fossil show? Describe how the place changed over time.

3. **Two-step problem.** The Supai Group is about 300 million years old and the Coconino Sandstone about 280 million. How much older is the Supai? Which one should be lower in the canyon?

4. **Find the mistake.** "A shark tooth is a trace fossil." Explain what is wrong and give a real example of a trace fossil.

★ **Challenge:** Draw two cliffs far apart with the same index fossil in one layer of each. Draw a line matching those layers and explain your match.

Looking Ahead: Next week is the Unit 4 Science Checkpoint, covering waves and light, energy resources, Earth's systems, and rock layers and fossils. Question to bring: which of Earth's spheres did coal come from, and which sphere is it in now?